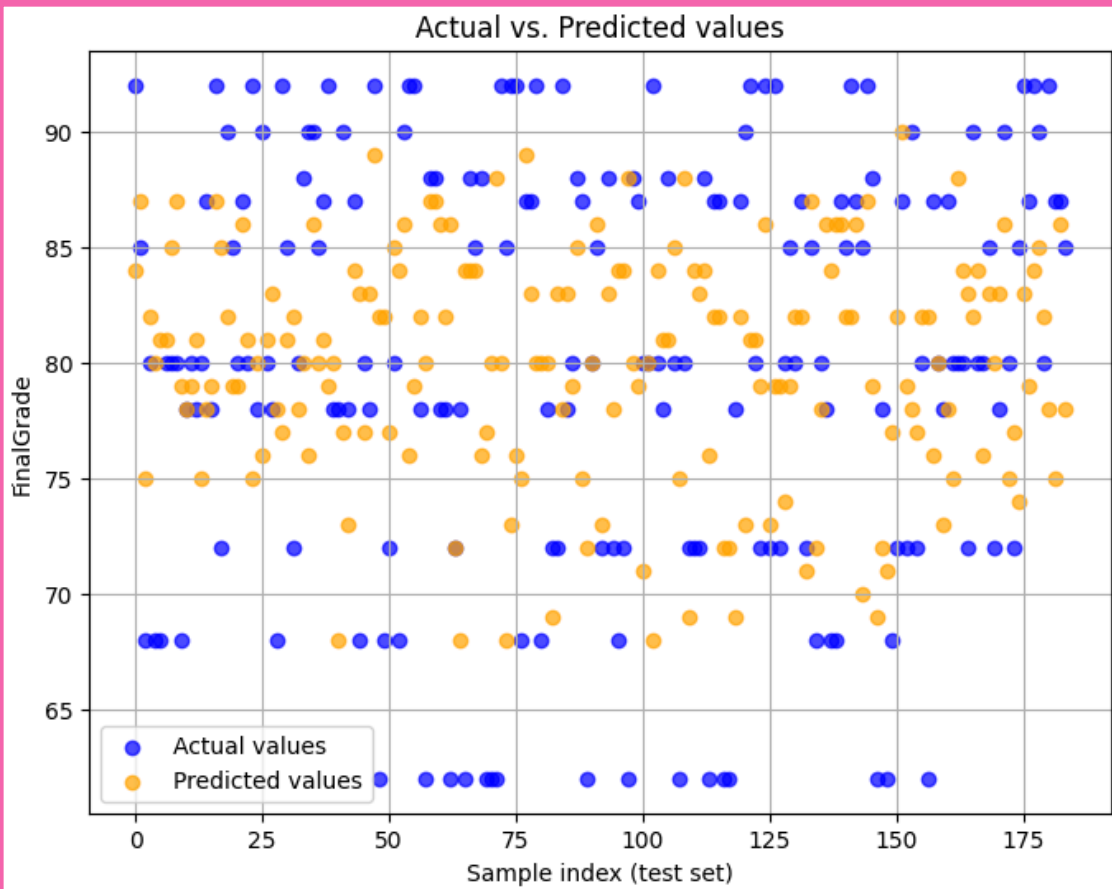


Student Grades Are Hard to Predict: A Machine Learning Analysis on Synthetic Educational Data



INTRODUCTION

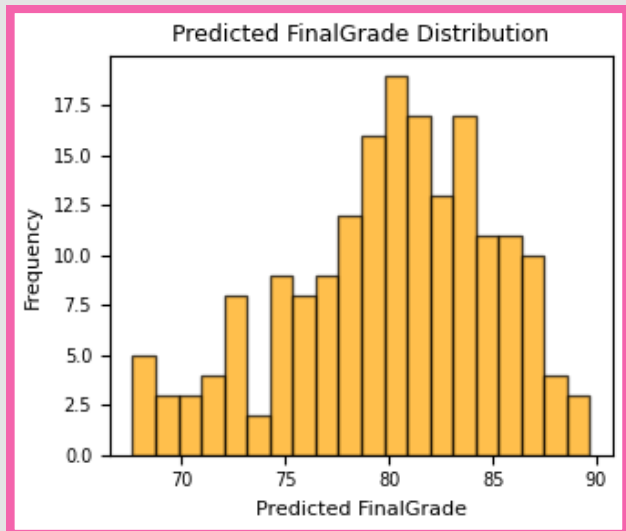
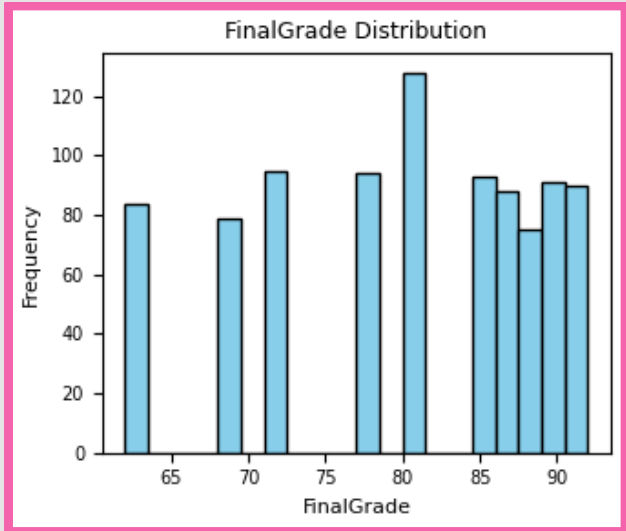
Predicting student performance is increasingly important for supporting early intervention and personalized learning. This study builds a complete data-processing and modeling pipeline using the synthetic Student Performance Predictions dataset from Kaggle [2]. The workflow includes data cleaning, outlier correction, missing value imputation, categorical encoding, and preparing training-test sets. A Random Forest regression [1] model is trained to estimate final grades, and its performance is evaluated with accuracy metrics, predicted-actual comparisons, and feature-importance analysis. Although the dataset contains realistic noise and intentionally flawed values, most features show only weak correlations with the target variable, making accurate prediction challenging.



CONCLUSION

This study developed a full data-processing and machine learning pipeline to predict students' final grades using a synthetic educational dataset. Despite extensive preprocessing and training a Random Forest regressor, the model showed weak performance, with high error and a negative R^2 score. These results indicate that the available features lack meaningful predictive power. The main limitation stems from the dataset itself, which was intentionally noisy and contains only weak correlations with the FinalGrade variable.

DISTRIBUTION



METODOLOGY

A Random Forest Regressor was chosen to predict students' final grades. As an ensemble of decision trees, it aggregates multiple models to improve stability and reduce overfitting. This approach is well suited for noisy, heterogeneous tabular data and can capture complex, non-linear relationships between features and academic outcomes.

REFERENCES

- [1] L. Breiman, "Random forests," Machine Learning, 45(1), 5–32, 2001.
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